Epistorm-mix

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Refs:

- https://github.com/epistorm/Epistorm-Mix/tree/main/matrices,
- https://www.epistorm.org/data/epistorm-mix
- https://github.com/epistorm/Epistorm-Mix/tree/main/data/imputed_data

```
xdir = "~/Documents/GitHub/Epistorm-Mix/tutorials/data-analysis/data/"
suppressMessages(library(tidyverse))
```

Lets do some random spot checks to see if this will work for our purposes

Person 1

Lets look at the contact patterns for 1 person

```
person_db <- read.csv(file.path(xdir, "imputed_resp.csv"))
subset(person_db, resp_id == 163925)
```

```
##   resp_id resp_age resp_gender resp_educator   weight
## 1  163925       67      Female            No 1.201913
```

a 67 year old Female

Looking in the contact patterns db

```
contact_db <- read.csv(file.path(xdir, "imputed_cont.csv"))
subset(contact_db, resp_id == 163925)
```

```
##   resp_id cont_id cont_age cont_setting           cont_ethn cont_gender
## 1  163925    1066       80    Household White, Non-Hispanic        Male
## 2  163925    1067       25        Other White, Non-Hispanic      Female
##                      cont_relationship
## 1 Person [My Child Lives/I Live] With
## 2                                Other
```

This persons *only* contact are:

- Household, 80yr Male
- Other, 25yr Female

Seems low?

Person 2

Lets look at the contact patterns for 1 person

```
subset(person_db, resp_id == 325231)
```

```
##   resp_id resp_age resp_gender resp_educator    weight
## 5  325231       15        Male            No 0.9484297
```

a 15 year old Male

Looking in the contact patterns db

```
subset(contact_db, resp_id == 325231)
```

```
##    resp_id cont_id cont_age cont_setting           cont_ethn cont_gender
## 12  325231    4154       50    Household     Hispanic/Latino        Male
## 13  325231    4155       46    Household     Hispanic/Latino      Female
## 14  325231    4156       28    Household     Hispanic/Latino        Male
## 15  325231    4157       15       School White, Non-Hispanic        Male
## 16  325231    4158       15       School White, Non-Hispanic        Male
## 17  325231    4159       14       School     Hispanic/Latino        Male
## 18  325231    4160       16       School     Hispanic/Latino        Male
## 19  325231    4161       15       School     Hispanic/Latino        Male
## 20  325231    4162       15       School White, Non-Hispanic        Male
##                      cont_relationship
## 12 Person [My Child Lives/I Live] With
## 13 Person [My Child Lives/I Live] With
## 14 Person [My Child Lives/I Live] With
## 15                  Friend/Acquaintance
## 16                  Friend/Acquaintance
## 17                  Friend/Acquaintance
## 18                  Friend/Acquaintance
## 19                     Other Schoolmate
## 20                    Classmate/Teacher
```

This persons *only* contact are:

- 3 household contacts
- and 6 school contacts
- and the teacher is 16

No contacts in "other" (the community setting)

and the school contacts seem low, both in other children and for teachers

Population level 1-10

```
pop_grp1 <- subset(person_db, resp_age %in% c(1:10))
all_contacts <- subset(contact_db, resp_id %in% c(pop_grp1$resp_id))
all_contacts %>%
  group_by(resp_id, cont_setting) %>%
  summarize(
    n = n()
  ) %>%
  ungroup() %>%
  group_by(cont_setting) %>%
  summarize(
    avg_n = mean(n)
  )
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by resp_id and cont_setting.
## i Output is grouped by resp_id.
```

```
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(resp_id, cont_setting))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.

## # A tibble: 4 x 2
##   cont_setting avg_n
##   <chr>        <dbl>
## 1 Household     3.10
## 2 Other         3.42
## 3 School        7.96
## 4 Work          1
```

Population level 60-65

```
pop_grp1 <- subset(person_db, resp_age %in% c(60:65))
all_contacts <- subset(contact_db, resp_id %in% c(pop_grp1$resp_id))
all_contacts %>%
  group_by(resp_id, cont_setting) %>%
  summarize(
    n = n()
  ) %>%
  ungroup() %>%
  group_by(cont_setting) %>%
  summarize(
    avg_n = mean(n)
  )
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by resp_id and cont_setting.
## i Output is grouped by resp_id.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(resp_id, cont_setting))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.

## # A tibble: 4 x 2
##   cont_setting avg_n
##   <chr>        <dbl>
## 1 Household     1.58
## 2 Other         3.83
## 3 Work          9.62
## 4 <NA>          1
```

Population level 10-18

```
pop_grp1 <- subset(person_db, resp_age %in% c(10:18))
all_contacts <- subset(contact_db, resp_id %in% c(pop_grp1$resp_id))
all_contacts %>%
  group_by(resp_id, cont_setting) %>%
  summarize(
    n = n()
  ) %>%
  ungroup() %>%
  group_by(cont_setting) %>%
  summarize(
```

```r
    avg_n = mean(n)
  )
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by resp_id and cont_setting.
## i Output is grouped by resp_id.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(resp_id, cont_setting))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```
## # A tibble: 5 x 2
##   cont_setting avg_n
##   <chr>        <dbl>
## 1 Household     2.91
## 2 Other         3.58
## 3 School        6.76
## 4 Work         11.7
## 5 <NA>          1
```